

Gestion de production électrique à EDF et symétries structurées en PLNE

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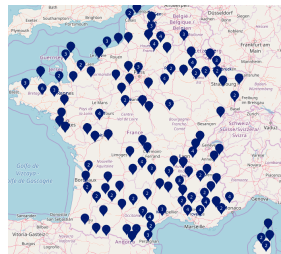
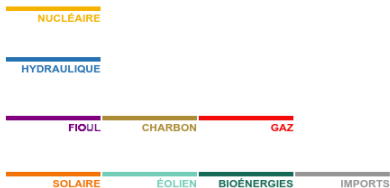
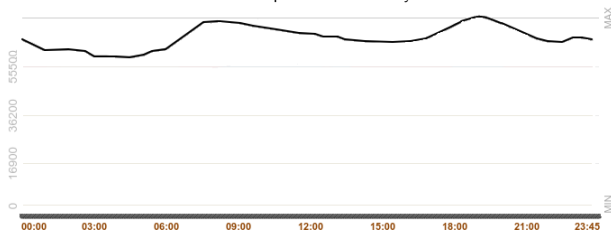
MAOA - 12 janvier 2022



- 1 The Unit Commitment Problem
 - Formal definition
 - Resolution at EDF
 - MILP formulation for the UCP
 - Lagrangian relaxation and Dantzig-Wolfe reformulation
- 2 Structured symmetries in Integer Linear Programs
 - Symmetry-breaking heuristics
 - Restrictions to representatives : using the full orbitope
 - Reformulations
- 3 Sub-symmetries in Integer Linear Programming
 - Restrictions to representatives : using the full (sub-)orbitope
 - Reformulation
- 4 Numerical illustration on the Unit Commitment Problem

The Unit Commitment Problem (UCP)

Demand for electric power on a one-day horizon

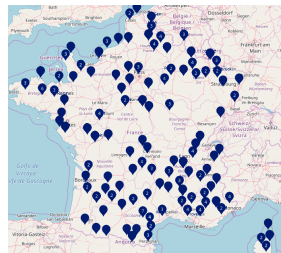
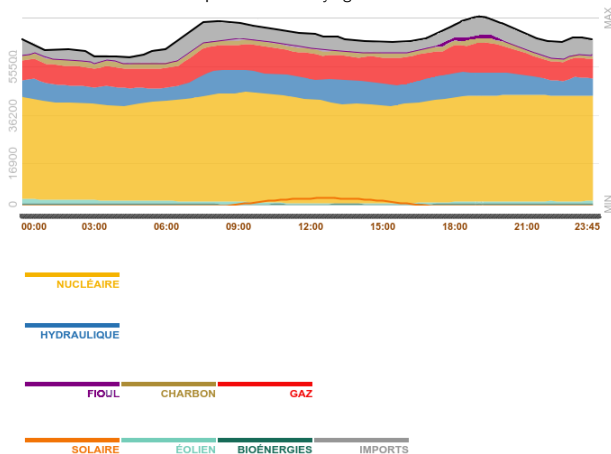


EDF power production units in France, including :

- 56 **nuclear** reactors
- 500 **hydro-power** plants dispatched in 50 valleys
- ■ ■ 20 **fossil-fuel** power plants
fuel oil/gas turbines
combined gas cycle
coal-fired power plants

The Unit Commitment Problem (UCP)

Power production satisfying the demand



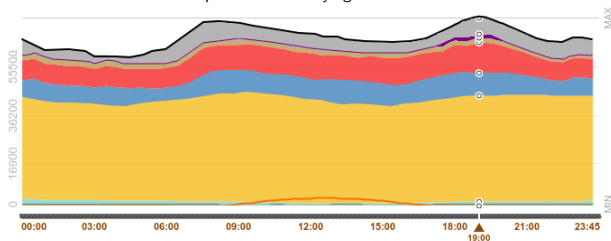
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→ Each production unit features technical constraints.

The Unit Commitment Problem (UCP)

Power production satisfying the demand

42135^{MW}

NUCLÉAIRE

5145^{MW}

HYDRAULIQUE

1598^{MW}

FIOUL

1753^{MW}

CHARBON

7890^{MW}

GAZ

0^{MW}

SOLAIRE

832^{MW}

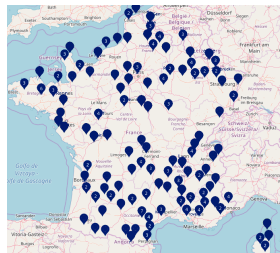
ÉOLIEN

1048^{MW}

BIOÉNERGIES

6444^{MW}

IMPORTS



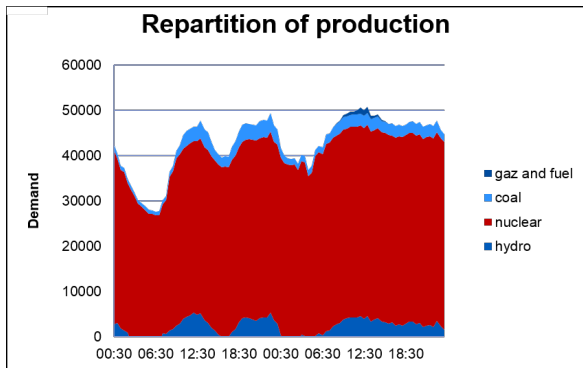
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Unit Commitment Problem at EDF

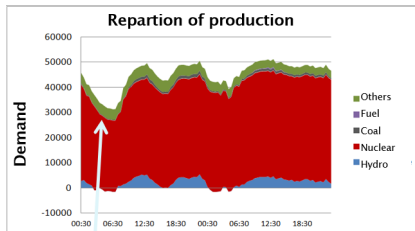
- **Demand consists of :**
 - Forecast of customer consumption (based on historical data, calendar, weather and temperature...)
 - Electricity bought or sold on the market
 - Substraction of the non dispatchable production forecast



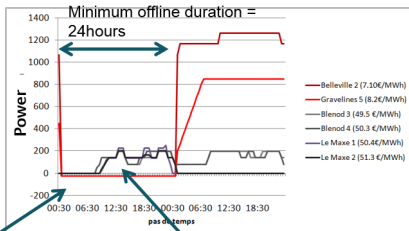
Importance of dynamic constraints

• Impact of dynamic constraints

- At each time step, not always possible to produce with the most economic production units
- When power demand varies quickly, not all production units are fit to modulate quickly enough



Very low demand on night



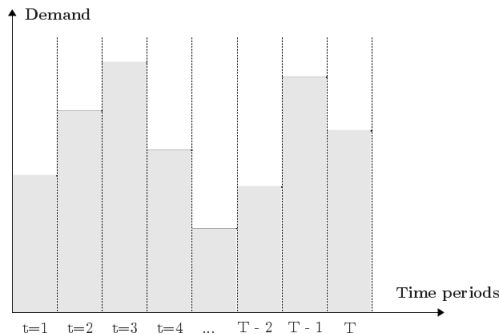
2 nuclear power plants offline

4 coal power plants started

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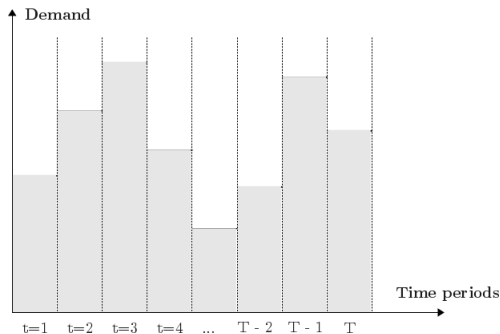
The Unit Commitment Problem (UCP)

- **Discretized** time horizon T
- **Demand** for electric power D_t at time t



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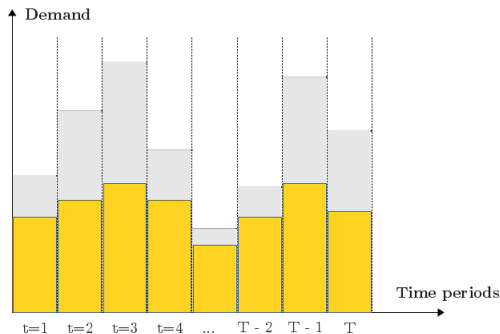


n production units



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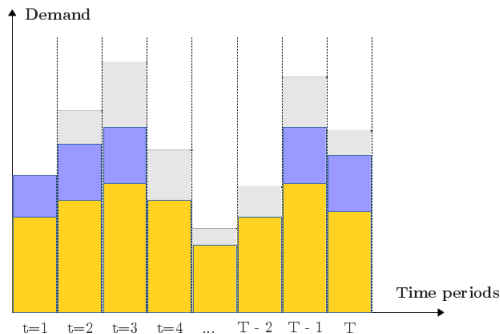


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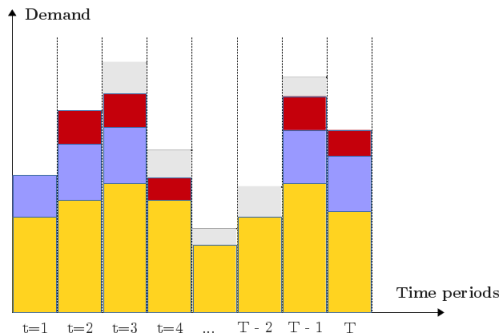


n production units



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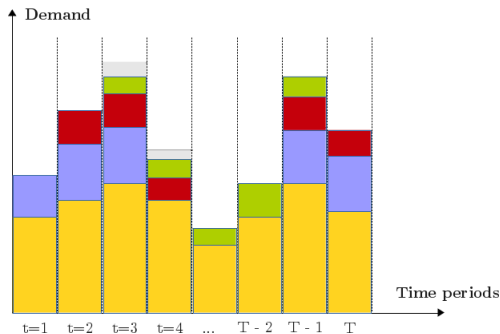


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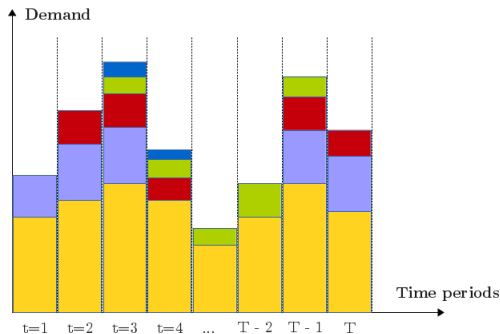


n production units



The Unit Commitment Problem (UCP)

- **Discretized** time horizon T
- **Demand** for electric power D_t at time t



n production units



The Unit Commitment Problem (UCP)

Production units must satisfy various technical constraints :

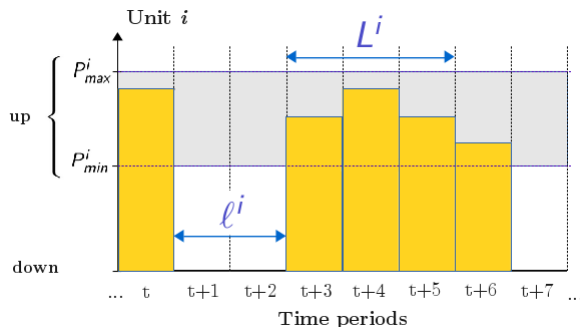
Thermal units (nuclear, gas, fuel oil, coal) :

- Minimum up and down times
- Production limits
- Ramp constraints, *i.e.*, production variation limits
- ...

Example : constraints of thermal units

Each thermal unit i must satisfy :

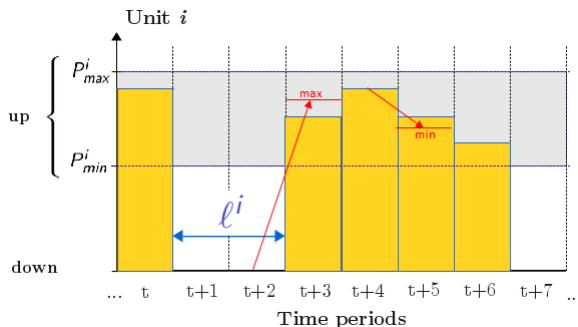
- Production limits P_{min}^i, P_{max}^i
- Minimum up time L^i
- Minimum down time ℓ^i
- Ramp constraints i.e. power variation is limited



Example : constraints of thermal units

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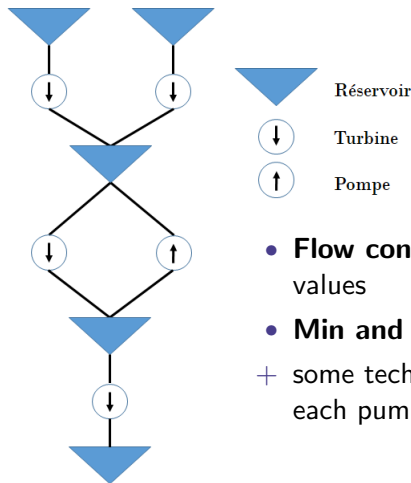
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The Unit Commitment Problem (UCP)

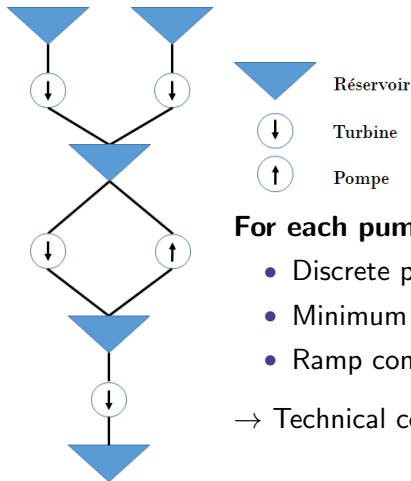
Constraints in hydro-valleys :

Hydro-valley = Network of turbines, pumps and reservoirs



- **Flow constraints** : conservation and min/max values
- **Min and max volume** in each reservoir
- + some technical constraints for each turbine and each pump

Technical constraints of pumps and turbines



For each pump and each turbine :

- Discrete power levels
- Minimum time on each power level
- Ramp constraints...

→ Technical constraints similar to thermal units.

Production costs

- **Thermal costs** for each thermal unit i :
 - **proportional** production cost c_p^i
 - **fixed** production cost c_f^i
 - **start-up** cost c_0^i
- **Hydraulic costs** :
 - **start-up** cost c_0^i for each hydro unit i
 - **Water values** computed with a mid-term vision (water remaining in reservoirs at the end of the horizon is seen as a profit in the objective)
→ 1 value per reservoir

Unit Commitment Problem (UCP)

Find a production plan that

- satisfies technical constraints of all units
- meets the **demand** at each time
- minimizes the total production cost

UCP resolutions at EDF

At EDF :

The UCP is solved for various time horizons :

- Mid-term horizons (from one year ahead to one week ahead)
- Day to day (for the next day)
→ **DAY-AHEAD UCP**
- Hour to hour (for the next hour to the end of the day)
→ **INTRA-DAY UCP**

The UCP is solved for various areas :

- Metropolitan France
- Insular regions (Corse, Guadeloupe, Réunion...) with grid constraints
→ **INSULAR UCP**

UCP resolutions at EDF

DAY-AHEAD UCP : Lagrangian relaxation of demand constraints

- **Decomposition**

Each production unit (nuclear, coal, gas, fuel-oil, hydro-valley) corresponds to a subproblem :

- **Thermal** subproblems solved by **dynamic programming**
- **Hydro-valley** subproblems solved by **Branch&Bound**

- **Coordination** of the subproblems

→ *Historic resolution technique at EDF*

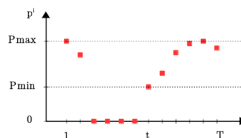
→ *Solves large scale instances with heterogeneous units on a daily basis*

INTRA-DAY UCP and **INSULAR SYSTEMS UCP** : **Branch&Bound**

MILP formulation

For each unit $i \in \{1, \dots, n\}$ and each time $t \in \{1, \dots, T\}$

- **production** variables $p_t^i \in \mathbb{R}$



$$\min_p \quad \sum_{i=1}^n \text{cost}(p^i)$$

$$\text{s. t.} \quad p^i \in \Pi^i \quad \forall i \in \{1, \dots, n\} \quad (1)$$

$$\sum_{i=1}^n p_t^i \geq D_t \quad \forall t \in \{1, \dots, T\} \quad (2)$$

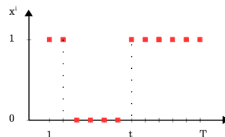
$$p_t^i \in \mathbb{R} \quad \forall i \in \{1, \dots, n\}, \quad \forall t \in \{1, \dots, T\}$$

- Operating domain of unit i : Π^i

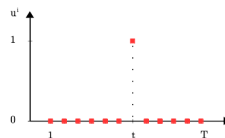
Example of an operating domain (thermal unit)

For each unit $i \in \{1, \dots, n\}$ and each time $t \in \{1, \dots, T\}$

- **up/down status** variables $x_t^i \in \{0, 1\}$



- **start-up** variables $u_t^i \in \{0, 1\}$



Example of an operating domain (thermal unit)

$$\Pi^i = \left\{ p \mid \begin{array}{ll} \sum_{t'=t-L^i+1}^t u_{t'}^i \leq x_t^i & \forall i \in \{1, \dots, n\}, \forall t \in \{L^i, \dots, T\} \\ \sum_{t'=t-\ell^i+1}^t u_{t'}^i \leq 1 - x_{t-\ell^i}^i & \forall i \in \{1, \dots, n\}, \forall t \in \{\ell^i + 1, \dots, T\} \\ u_t^i \geq x_t^i - x_{t-1}^i & \forall i \in \{1, \dots, n\}, \forall t \in \{2, \dots, T\} \\ P_{min}^i x_t^i \leq p_t^i \leq P_{max}^i x_t^i & \forall i \in \{1, \dots, n\}, \forall t \in \{1, \dots, T\} \\ u_t^i, x_t^i \in \{0, 1\}, p_t^i \in \mathbb{R} & \forall i \in \{1, \dots, n\}, \forall t \in \{1, \dots, T\} \end{array} \right\}$$

Lagrangian relaxation

$$\begin{aligned}
 \min_p \quad & \sum_{i=1}^n \text{cost}(p^i) \\
 \text{s. t.} \quad & p^i \in \Pi^i \quad \forall i \in \{1, \dots, n\} \\
 & \sum_{i=1}^n p_t^i \geq D_t \quad \forall t \in \{1, \dots, T\} \\
 & p_t^i \in \mathbb{R} \quad \forall i \in \{1, \dots, n\}, \forall t \in \{1, \dots, T\}
 \end{aligned}$$



$$\begin{aligned}
 \min_p \quad & \sum_{i=1}^n \text{cost}(p^i) + \sum_{t=1}^T \lambda_t (D_t - \sum_{i=1}^n p_t^i) \\
 \text{s. t.} \quad & p^i \in \Pi^i \quad \forall i \in \{1, \dots, n\} \\
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 \end{aligned}$$



$$\begin{aligned}
 \min_p \quad & \sum_{i=1}^n \left(cost(p^i) - \sum_{t=1}^T \lambda_t p_t^i \right) \\
 \text{s. t.} \quad & p^i \in \Pi^i \quad \forall i \in \{1, \dots, n\} \\
 & p_t^i \in \mathbb{R} \quad \forall i \in \{1, \dots, n\}, \forall t \in \{1, \dots, T\}
 \end{aligned}$$

Decomposition in subproblems

$$\begin{aligned}
 \min_p \quad & \sum_{i=1}^n \left(\text{cost}(p^i) - \sum_{t=1}^T \lambda_t p_t^i \right) \\
 \text{s. t.} \quad & p^i \in \Pi^i \quad \forall i \in \{1, \dots, n\} \\
 & p_t^i \in \mathbb{R} \quad \forall i \in \{1, \dots, n\}, \forall t \in \{1, \dots, T\}
 \end{aligned}$$

$$\downarrow$$

$$\dots$$

$$\downarrow$$

$$\begin{aligned}
 \min_p \quad & \left(\text{cost}(p^1) - \sum_{t=1}^T \lambda_t p_t^1 \right) \\
 \text{s. t.} \quad & p^1 \in \Pi^1 \\
 & p_t^1 \in \mathbb{R} \quad \forall t \in \{1, \dots, T\}
 \end{aligned}$$

$$\begin{aligned}
 \min_p \quad & \left(\text{cost}(p^n) - \sum_{t=1}^T \lambda_t p_t^n \right) \\
 \text{s. t.} \quad & p^n \in \Pi^n \\
 & p_t^n \in \mathbb{R} \quad \forall t \in \{1, \dots, T\}
 \end{aligned}$$

→ n subproblems : 1 per unit

Dantzig-Wolfe reformulation

For each unit i , Π^i set of feasible production plans.

New variable space : for each $\pi \in \Pi^i$,

$\mu_\pi = 1$ iff **plan π chosen for unit i**

$$\begin{aligned}
 (DW_{UCP}) \quad & \min_{\mu} \quad \sum_{i=1}^n \sum_{\pi \in \Pi^i} c^\pi \mu^\pi \\
 \text{s. t.} \quad & \sum_{\pi \in \Pi^i} \mu^\pi = 1 && \forall i \in \{1, \dots, n\} \\
 & \sum_{i=1}^n \sum_{\pi \in \Pi^i} p_t^\pi \mu^\pi \geq D_t && \forall t \in \{1, \dots, T\} \\
 & \mu^\pi \in \{0, 1\} && \forall i \in \{1, \dots, n\} \quad \forall \pi \in \Pi^i
 \end{aligned}$$

p_t^π : power produced by unit i at time t in plan $\pi \in \Pi^i$

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- Symmetry-breaking heuristics
- Restrictions to representatives : using the full orbitope
- Reformulations

3 Sub-symmetries in Integer Linear Programming

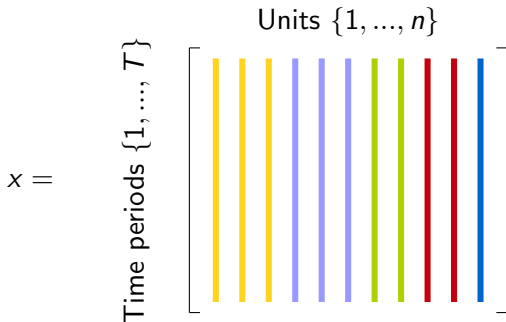
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- Reformulation

4 Numerical illustration on the Unit Commitment Problem

Symmetries in the UCP

→ **Symmetries in the UCP** come from the existence of **identical units**.

- Variables $x_t^i \in \{0, 1\}$: unit i up at time t .
→ matrix $x = (x_t^i)$
- **Groups of identical units**, i.e., units with same characteristics & costs



Example : 3 identical units with horizon $T = 3$

→ given one solution $x_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

→ 5 symmetric solutions :

$$x_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$x_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$x_5 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$x_6 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Symmetry in integer programming - Definitions

$$\min \left\{ c(x) \mid x \in \mathcal{X} \right\} \quad (ILP)$$

$\mathcal{X}$ subset of $m \times n$ binary matrices

- A **symmetry** is a permutation π of the variables s.t.

$$\forall x \in \mathcal{X}, \pi(x) \in \mathcal{X} \text{ and } c(x) = c(\pi(x))$$

$$x = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad x' = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

- The **symmetry group** $\mathcal{G}$ of (ILP) is the set of all symmetries.
- Symmetry group $\mathcal{G}$ partitions $\mathcal{X}$ into **orbits**.

In this presentation...

We consider problems with **structured symmetries**, *i.e.*

any column permutation is a symmetry.

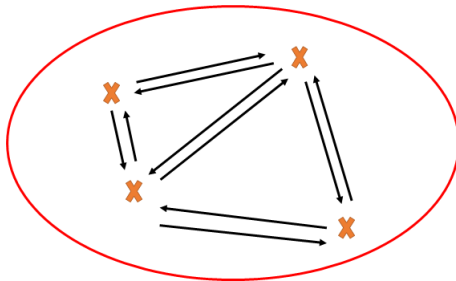
Thus, the symmetry group $\mathcal{G}$ is the set of all column permutations.

In the UCP, not exactly **all** column permutations but :

$$x = \begin{array}{c} \text{Units } \{1, \dots, n\} \\ \left[\begin{array}{cccccccccccc} \text{yellow} & \text{yellow} & \text{yellow} & \text{blue} & \text{blue} & \text{blue} & \text{green} & \text{green} & \text{green} & \text{red} & \text{red} & \text{blue} \end{array} \right] \end{array}$$

→ everything we do in the "all column permutation" case will also apply to this more general case.

Orbits



Orbits : for any solution x , the orbit of x contains :

every solution obtained by arbitrarily permuting x 's columns

Example : 3 identical units with horizon $T = 3$

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$$x_6 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

→ **orbit of x_1 = orbit of x_2 = ... = orbit of x_6 = $\{x_1, x_2, x_3, x_4, x_5, x_6\}$**

Example : 3 identical units with horizon $T = 3$

→ for another solution $y_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

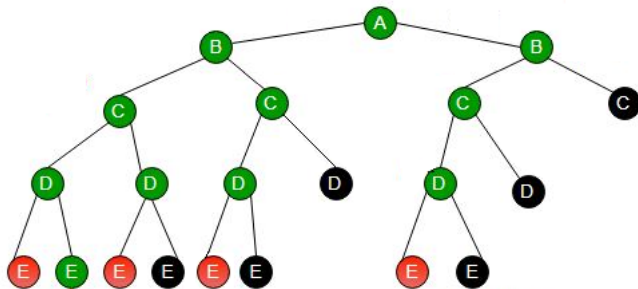
→ 2 symmetric solutions :

$$y_2 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \qquad y_3 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

→ **orbit of** y_1 = **orbit of** y_2 = **orbit of** y_3 = $\{y_1, y_2, y_3\}$

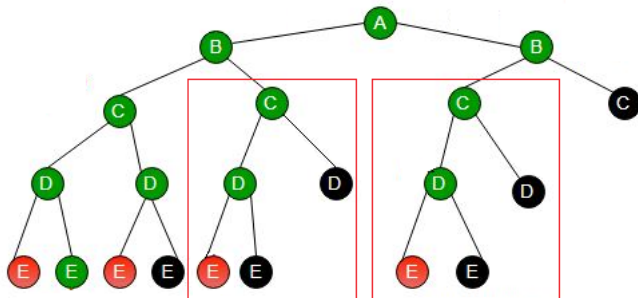
Symmetries in ILP

Many symmetric solutions lead to many symmetric branches in the B&B tree that cannot be pruned.



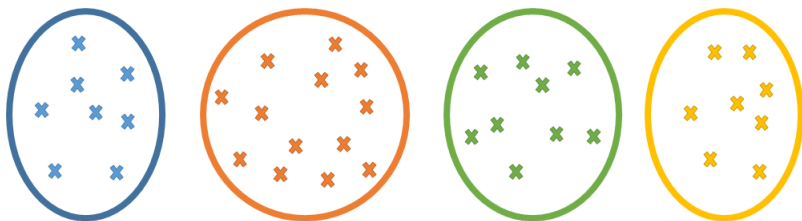
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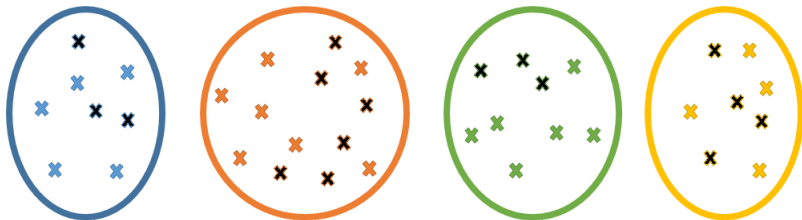
→ the B&B tree might get very big !

How to break symmetries ?



How to restrict the search space to non-symmetric solutions only ?

How to break symmetries ?

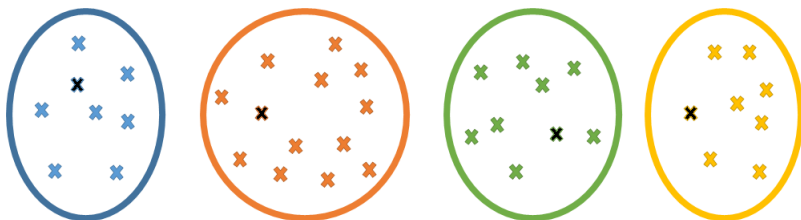


How to restrict the search space to non-symmetric solutions only ?

- Use symmetry-breaking **heuristics** removing some symmetric solutions (but not all)

NB : black crosses = solutions not removed

How to break symmetries ?



How to restrict the search space to non-symmetric solutions only ?

- Use symmetry-breaking **heuristics** removing some symmetric solutions (but not all)
- **Choose one representative** per orbit and **restrict** the search to the representative set

How to break symmetries ?



How to restrict the search space to non-symmetric solutions only ?

- Use symmetry-breaking **heuristics** removing some symmetric solutions (but not all)
- **Choose one representative** per orbit and **restrict** the search to the representative set
- **Reformulate** (i.e., change variable space)
→ indeed, symmetries depend on how the problem is formulated

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1-counting inequalities

Idea : restrict the search to solutions such that :

number of 1's in column $j \geq$ number of 1's in column $j + 1$

$$\sum_{i=1}^m x_{i,j} \geq \sum_{i=1}^m x_{i,j+1}, \quad \forall j \in \{1, \dots, n-1\}$$

(Lima and Novais, 2016)

Example : 3 identical units with horizon $T = 3$

→ 6 symmetric solutions :

$$x_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad \rightarrow \text{these solutions remain}$$

$$x_3 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad x_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \rightarrow \text{these solutions are removed}$$

$$x_5 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad x_6 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad \rightarrow \text{these solutions are removed}$$

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Orbital branching

At node a of the B&B tree : instead of branching on $x_{i,j} = 0 \vee x_{i,j} = 1$

- look for variables symmetric to $x_{i,j}$ at node a

At node a , some variables are fixed :

1	×	1	1	1
×	0	×	×	×
×	1	×	×	×
0	1	×	0	0

with arbitrary integer α

(Ostrowski, Anjos and Vannelli, 2015)

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At node a , some variables are fixed :

1	×	1	1	1
×	0	×	×	×
×	1	×	×	×
0	1	×	0	0

Set of symmetric variables $O = \{x_{i,j_1}, x_{i,j_2}, \dots, x_{i,j_k}\}$

- branch on :**

at least α variables of O are equal to 1

$\vee$

at most $\alpha - 1$ variables of O are equal to 1

with arbitrary integer α

(Ostrowski, Anjos and Vannelli, 2015)

Orbital branching

at least α variables of O are equal to 1

$\vee$

at most $\alpha - 1$ variables of O are equal to 1

$\downarrow$

at least α variables of O are equal to 1

$\vee$

at least $k - \alpha$ var. of O equal 0

$\downarrow$

(By symmetry)

first α variables of O are equal to 1

$\vee$

last $k - \alpha$ variables of O are equal to 0

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Lexicographic order

→ **Order on the columns.**

Consider two columns $X, Y \in \{0, 1\}^m$.

Lexicographic order

Y is **lexicographically greater than** X (denoted $Y \succeq X$) if :

$\exists i \in \{1, \dots, m-1\}$ s. t.

- $\forall i' \leq i, Y_{i'} = X_{i'}$
- $Y_{i+1} = 1$ and $X_{i+1} = 0$

Example

$$Y = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \succeq X = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Full Orbitopes

- When the symmetry group is the set of all column permutations :

$x \in \mathcal{X}$ lex-order representative $\iff x(1) \succeq \dots \succeq x(n)$

$\iff$ **Columns of x are in non-increasing lexicographic order**

Example

$x_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ is a lex-order representative.

Full Orbitope

$$\mathcal{P}_0 = \text{Conv} \left\{ \text{binary matrices } x \mid x(1) \succeq \dots \succeq x(n) \right\}$$

Solving on $\mathcal{X} \iff$ Solving on $\mathcal{X} \cap \mathcal{P}_0$

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First idea to enforce that each x is in the full orbitope :

add new constraints in the model

→ For a given matrix $x = (x_{i,j}) = [x(1), x(2), \dots, x(n)]$,
where $x(1), x(2), \dots, x(n)$ are x 's columns.

Find linear inequalities enforcing that : $x(1) \succeq x(2) \succeq \dots \succeq x(n)$

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where $x(1), x(2), \dots, x(n)$ are x 's columns.

Find linear inequalities enforcing that : $x(1) \succeq x(2) \succeq \dots \succeq x(n)$

One option : for each pair of columns $j, j+1$, add inequalities

$$2^{m-1}x_{1,j} + 2^{m-2}x_{2,j} + \dots + x_{m,j} \geq 2^{m-1}x_{1,j+1} + 2^{m-2}x_{2,j+1} + \dots + x_{m,j+1}$$

j	$j+1$	
$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$	→ weight 2^{m-1}
		→ weight 2^{m-2}
		...
		→ weight 2^0

→ **numerical difficulties** (exponential coefficients)

Full orbitope inequalities

Find linear inequalities enforcing that : $x(1) \succeq x(2) \succeq \dots \succeq x(n)$

Other option : add new variables

$$y_{i,j} = 1 \quad \text{iff} \quad \text{column } j \text{ and } j+1 \text{ are equal from row } 1 \text{ to } i$$

Example

	j	$j+1$	
$i = 1$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$y_{1,j} = 1$
$i = 2$	$\begin{bmatrix} 0 \end{bmatrix}$	$\begin{bmatrix} 0 \end{bmatrix}$	$y_{2,j} = 1$
$i = 3$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} 0 \end{bmatrix}$	$y_{3,j} = 0$
$i = 4$	$\begin{bmatrix} 0 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$y_{4,j} = 0$
$i = 5$	$\begin{bmatrix} 0 \end{bmatrix}$	$\begin{bmatrix} 0 \end{bmatrix}$	$y_{5,j} = 0$

(Loos, 2011)

(Bendotti, Fouilhoux and R., 2019)

Full orbitope inequalities

$y_{i,j} = 1$ iff column j and $j + 1$ are equal from row 1 to i

New constraints :

- $y_{i,j} = 1 \iff \begin{cases} y_{i-1,j} = 1 \\ x_{i,j} = x_{i,j+1} \end{cases}$
- If $\begin{cases} y_{i-1,j} = 1 \\ y_{i,j} = 0 \end{cases}$ then $\begin{cases} x_{i,j} = 1 \\ x_{i,j+1} = 0 \end{cases}$

Example

	j	$j + 1$	
$i = 1$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$y_{1,j} = 1$
$i = 2$	$\begin{bmatrix} 0 \end{bmatrix}$	$\begin{bmatrix} 0 \end{bmatrix}$	$y_{2,j} = 1$
$i = 3$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} 0 \end{bmatrix}$	$y_{3,j} = 0$
$i = 4$	$\begin{bmatrix} 0 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$y_{4,j} = 0$
$i = 5$	$\begin{bmatrix} 0 \end{bmatrix}$	$\begin{bmatrix} 0 \end{bmatrix}$	$y_{5,j} = 0$

Full orbitope inequalities

$$y_{i,j} = 1 \quad \text{iff} \quad \text{column } j \text{ and } j+1 \text{ are equal from row } 1 \text{ to } i$$

Constraints :

$$1 - y_{1,j-1} = x_{1,j-1} - x_{1,j}$$

$$y_{i-1,j-1} \leq y_{i,j-1} + x_{i,j-1}$$

$$y_{i-1,j-1} + x_{i,j} \leq 1 + y_{i,j-1}$$

$$x_{i,j-1} \leq 1 - y_{i,j-1} + x_{i,j}$$

$$y_{i,j-1} \leq y_{i-1,j-1}$$

$$x_{i,j} \leq (1 - 2y_{i-1,j} - y_{i,j}) + x_{i,j-1}$$

$$\forall j \in \{2, \dots, n\}$$

$$\forall i \in \{2, \dots, m\}, j \in \{2, \dots, n\}$$

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$$\forall i \in \{2, \dots, m\}, j \in \{2, \dots, n\}$$

→ complete linear description in variable space (x, y)
of the 2-column full orbitope

(Loos, 2011)

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Fixing in the full orbitope

Another idea to enforce that each x is in the full orbitope :

**fix variables during the B&B search
to ensure columns are in lex order**

- At each node a of the B&B tree, some variables are already fixed to 0 (variables I_0^a) or already fixed to 1 (variables I_1^a)
→ defines a set

$$F = \{ \text{matrices } x \mid x_{i,j} = 1 \ \forall (i,j) \in I_1^a \quad x_{i,j} = 0 \ \forall (i,j) \in I_0^a \}$$

- At node a , we want to enumerate solutions x such that

$$x \in F \cap P \cap \{0,1\}^{(m,n)}$$

where P is the full orbitope.

→ **it may be possible to fix more variables to 0 or to 1 at node a**

Example

At node a , there are some fixed variables :

("×" stands for free variables i.e. variables not fixed yet)

$$x = \begin{bmatrix} \times & 1 \\ \times & 1 \\ 0 & \times \end{bmatrix}$$

We want x 's columns to be in lexicographic order.

Example

At node a , there are some fixed variables :

("×" stands for free variables i.e. variables not fixed yet)

$$x = \begin{bmatrix} 1 & 1 \\ \times & 1 \\ 0 & \times \end{bmatrix}$$

We want x 's columns to be in lexicographic order.

Example

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We want x 's columns to be in lexicographic order.

Example

At node a , there are some fixed variables :

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$$x = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$$

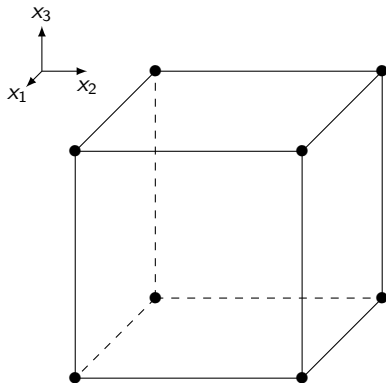
We want x 's columns to be in lexicographic order.

Variable fixing - Intersection with a cube face

V. Kaibel, M. Peinhardt, M. Pfetsch. *Orbitopal fixing*, Proc. IPCO, 2007.

- Let hypercube $C = \text{Conv} \{ m \times n \text{ binary matrices} \}$
- A non-empty face F of C is given by two index sets I_0, I_1 :

$$F = \{x \in C \mid x_{i,j} = 0 \ \forall (i,j) \in I_0, \quad x_{i,j} = 1 \ \forall (i,j) \in I_1\}.$$

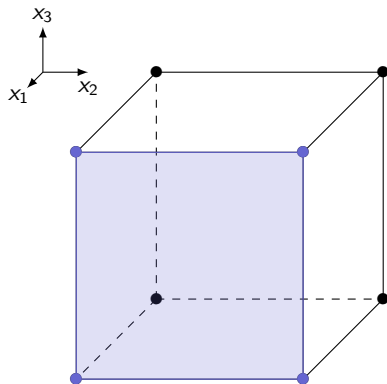


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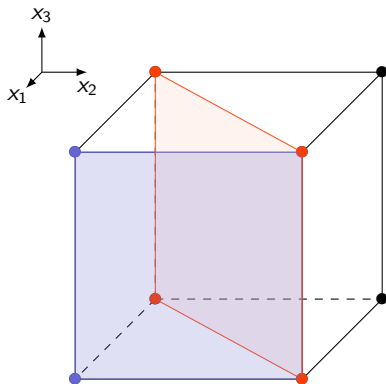
- For a given face F of C
 $F = \{x \in [0, 1]^3 \mid x_1 = 1\}$
 $I_1 = \{1\}, I_0 = \emptyset$

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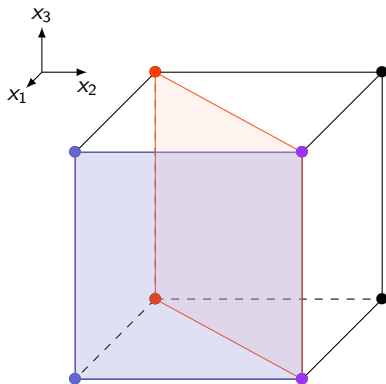
- For a given face F of C
- For a given polytope $P \subset C$

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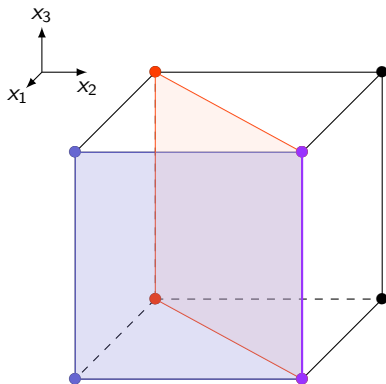
- For a given face F of C
 - For a given polytope $P \subset C$
- Solution set S
 $S = P \cap F \cap \{0, 1\}^{(m,n)}$

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- For a given face F of C
- For a given polytope $P \subset C$
 $\rightarrow$ Solution set S
 $S = P \cap F \cap \{0, 1\}^{(m,n)}$

Fixing of P at F

$\text{Fix}_F(P)$ = the smallest face of C that contains S

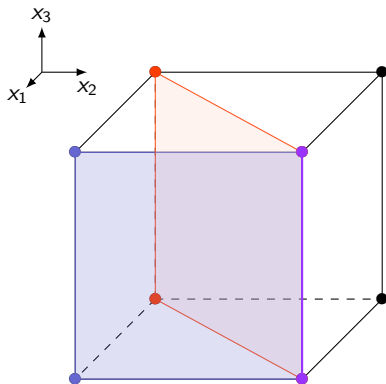
I_0^* and I_1^* : index sets defining $\text{Fix}_F(P)$ (if non-empty)

Variable fixing - Intersection with a cube face

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- For a given face F of C
- For a given polytope $P \subset C$
 $\rightarrow$ Solution set S
 $S = P \cap F \cap \{0, 1\}^{(m,n)}$

Fixing of P at F

$\text{Fix}_F(P)$ = the smallest face of C that contains S

Fixing $\text{Fix}_F(P)$: $I_0^* = \emptyset$,
 $I_1^* = \{1, 2\}$, i.e., $x_1 = x_2 = 1$

Orbitopal fixing in the B&B tree

V. Kaibel, M. Peinhardt, M. Pfetsch. *Orbitopal fixing*, Proc. IPCO, 2007.

We want to restrict the solution set $\mathcal{X}$ to a polytope P ,

→ **we want to enumerate only solutions in $\mathcal{X} \cap P$.**

Orbitopal fixing algorithm

At each node a of the B&B tree :

- (INPUT) I_0^a : variables set to 0, I_1^a : variables set to 1
→ F cube face defined by I_0^a, I_1^a
- **Compute** I_0^* and I_1^* : index sets defining $\text{Fix}_F(P)$
- (OUTPUT) Fixed variables :
→ **Fix** to 0 variables in I_0^* and to 1 variables in I_1^*

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- (OUTPUT) Fixed variables :
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Orbitopal fixing for the full orbitope

Let $P = \mathcal{P}_0$ (the full orbitope).

For a given face F :

- $\text{Fix}_F(P)$: smallest cube face containing the intersection $F \cap P \cap \{0, 1\}^{(m,n)}$.
- I_0^* and I_1^* : index sets defining $\text{Fix}_F(P)$

Theorem

$\forall F, I_0^$ and I_1^* can be computed in linear time.*

Idea :

Find binary matrices $M_{\min}$ and $M_{\max}$ in $F \cap P$ s.t.

$\forall x \in F \cap P \cap \{0, 1\}^{(m,n)} :$

$$\text{i.e. } \forall \text{ column } j, \quad M_{\min}(j) \preceq x(j) \preceq M_{\max}(j)$$

(Bendotti, Fouilhoux and R., 2018)

Computing I_0^* and I_1^* - Example

Example : face F defined by $I_0 = \{(4, 2), (3, 3), (5, 3)\}$ and $I_1 = \{(1, 1), (2, 1), (4, 1), (2, 2), (5, 2), (4, 3), (1, 4), (2, 4)\}$

$\forall x \in F \cap P \cap \{0, 1\}^{(m,n)}$, x of the form

$$\begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \times \\ 1 & 0 & 1 & \times \\ \times & 1 & 0 & \times \end{bmatrix}$$

$$M_F$$

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$\forall x \in F \cap P \cap \{0, 1\}^{(m,n)}$, x of the form

$$\begin{array}{ccc}
 \begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \times \\ 1 & 0 & 1 & \times \\ \times & 1 & 0 & \times \end{bmatrix} & \preceq & \begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \times \\ 1 & 0 & 1 & \times \\ \times & 1 & 0 & \times \end{bmatrix} \\
 M_{min} & & M_F \\
 & & \preceq & \begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \times \\ 1 & 0 & 1 & \times \\ \times & 1 & 0 & \times \end{bmatrix} \\
 & & & M_{max}
 \end{array}$$

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$\forall x \in F \cap P \cap \{0, 1\}^{(m,n)}$, x of the form

$$\begin{array}{ccc}
 \begin{bmatrix} 1 & \times & \times & \textcolor{red}{1} \\ 1 & 1 & \times & \textcolor{red}{1} \\ \times & \times & 0 & \textcolor{red}{\times} \\ 1 & 0 & 1 & \textcolor{red}{\times} \\ \times & 1 & 0 & \textcolor{red}{\times} \end{bmatrix} & \preceq & \begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \times \\ 1 & 0 & 1 & \times \\ \times & 1 & 0 & \times \end{bmatrix} \\
 M_{min} & & M_F \\
 & \preceq & \begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \times \\ 1 & 0 & 1 & \times \\ \times & 1 & 0 & \times \end{bmatrix} \\
 & & M_{max}
 \end{array}$$

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Example : face F defined by $I_0 = \{(4, 2), (3, 3), (5, 3)\}$ and $I_1 = \{(1, 1), (2, 1), (4, 1), (2, 2), (5, 2), (4, 3), (1, 4), (2, 4)\}$

$\forall x \in F \cap P \cap \{0, 1\}^{(m,n)}$, x of the form

$$\begin{array}{ccc}
 \begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \color{red}{0} \\ 1 & 0 & 1 & \color{red}{0} \\ \times & 1 & 0 & \color{red}{0} \end{bmatrix} & \preceq & \begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \times \\ 1 & 0 & 1 & \times \\ \times & 1 & 0 & \times \end{bmatrix} \\
 M_{min} & & M_F \\
 & & \preceq & \begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \times \\ 1 & 0 & 1 & \times \\ \times & 1 & 0 & \times \end{bmatrix} \\
 & & & M_{max}
 \end{array}$$

Computing I_0^* and I_1^* - Example

Example : face F defined by $I_0 = \{(4, 2), (3, 3), (5, 3)\}$ and $I_1 = \{(1, 1), (2, 1), (4, 1), (2, 2), (5, 2), (4, 3), (1, 4), (2, 4)\}$

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 & & \preceq & \begin{bmatrix} 1 & \times & \times & 1 \\ 1 & 1 & \times & 1 \\ \times & \times & 0 & \times \\ 1 & 0 & 1 & \times \\ \times & 1 & 0 & \times \end{bmatrix} \\
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$$\Rightarrow I_0^* \text{ and } I_1^*$$

1 The Unit Commitment Problem

- Formal definition
- Resolution at EDF
- MILP formulation for the UCP
- Lagrangian relaxation and Dantzig-Wolfe reformulation

2 Structured symmetries in Integer Linear Programs

- Symmetry-breaking heuristics
 - 1-counting inequalities
 - Orbital branching
- Restrictions to representatives : using the full orbitope
 - Full orbitope inequalities
 - Fixing in the full orbitope
- Reformulations
 - Dantzig-Wolfe reformulation
 - Aggregation of symmetric variables

3 Sub-symmetries in Integer Linear Programming

- Restrictions to representatives : using the full (sub-)orbitope
 - Orbitopal fixing
 - Sub-symmetry-breaking inequalities
- Reformulation

4 Numerical illustration on the Unit Commitment Problem

Dantzig-Wolfe reformulation : UCP example

For each unit i , Π^i set of feasible production plans.

New variable space : for each $\pi \in \Pi^i$,

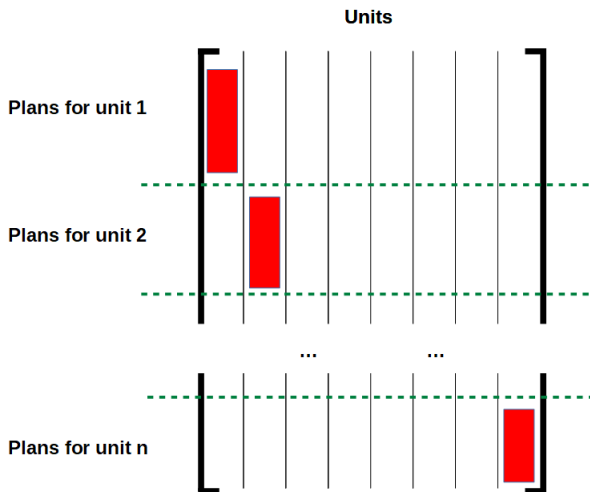
$\mu_\pi = 1$ iff **plan π chosen for unit i**

$$\begin{aligned}
 (DW_{UCP}) \quad & \min_{\mu} \quad \sum_{i=1}^n \sum_{\pi \in \Pi^i} c^\pi \mu^\pi \\
 \text{s. t.} \quad & \sum_{\pi \in \Pi^i} \mu^\pi = 1 && \forall i \in \{1, \dots, n\} \\
 & \sum_{i=1}^n \sum_{\pi \in \Pi^i} p_t^\pi \mu^\pi \geq D_t && \forall t \in \{1, \dots, T\} \\
 & \mu^\pi \in \{0, 1\} && \forall i \in \{1, \dots, n\} \quad \forall \pi \in \Pi^i
 \end{aligned}$$

p_t^π : power produced by unit i at time t in plan $\pi \in \Pi^i$

Dantzig-Wolfe reformulation with identical units

→ less symmetric solution space !



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Aggregation of symmetric variables

$$x = \begin{bmatrix} \text{12 columns of colored bars} \end{bmatrix}$$

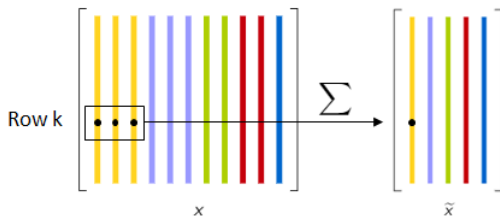
For set J of “permutable” columns, each row k , **define aggregated variables** :

$$\tilde{x}_{k,J} = \sum_{j \in J} x_{k,j}$$

$\tilde{x}_{k,J}$ = number of variables to 1 among variables $x_{k,j}$, $j \in J$

(Fischetti, Liberti, Salvagnin and Walsh, 2017)

Aggregation of symmetric variables



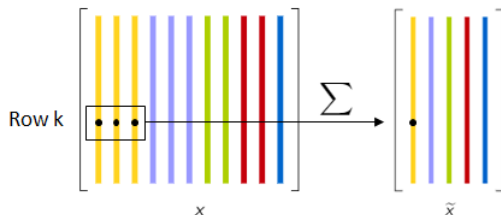
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$\tilde{x}_{k,J}$ = number of variables to 1 among variables $x_{k,j}$, $j \in J$

Replace every occurrence of variable $x_{k,j}$ by $\frac{\tilde{x}_{k,J}}{|J|}$
in the integer program, for each $j \in J$.

(Fischetti, Liberti, Salvagnin and Walsh, 2017)

$PC = \{ \text{set of permutable columns} \}$

Original ILP

$$\begin{aligned}
 (P) \quad & v = \min_x \sum_{k=1}^m \sum_{j=1}^n c_{k,j} x_{k,j} \\
 \text{s. t.} \quad & Ax \leq b \\
 & x_{k,j} \in \{0, 1\} \quad \forall j \in \{1, \dots, n\}, \forall k \in \{1, \dots, m\}
 \end{aligned}$$

↓ (relaxation)

Agregated ILP

$$\begin{aligned}
 (\tilde{P}) \quad & \tilde{v} = \min_{\tilde{x}} \sum_{k=1}^m \sum_{J \in PC} \tilde{c}_{k,J} \tilde{x}_{k,J} \\
 \text{s. t.} \quad & \tilde{A} \tilde{x} \leq b \\
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 & \tilde{x}_{k,J} \in \{0, 1, \dots, |J|\} \quad \forall J \in PC, \forall k \in \{1, \dots, m\}
 \end{aligned}$$

Reduces the number of variables

But replaces binary variables by integer variables

(Fischetti, Liberti, Salvagnin and Walsh, 2017)

General case

In general : $(\tilde{P})$ is a relaxation of (P) , i.e. $\tilde{v} \leq v$.

Indeed, for a given solution $\tilde{x}$, there may not exist x such that $\tilde{x}_{k,J} = \sum_{j \in J} x_{k,j}$ for each J and k , and $Ax \leq b$, $x_{kj} \in \{0, 1\}$

(Fischetti, Liberti, Salvagnin and Walsh, 2017)

But in some particular cases, this is an exact reformulation : $\tilde{v} = v$

→ **Depends on constraint formulations :**

- **If constraint matrix A is totally unimodular (TU) :**

for any solution $\tilde{x} \in \mathbb{Z}^n$ and $k \in \mathbb{N}$ such that $A\tilde{x} \leq kb$, there exists

$x_1, \dots, x_k \in \mathbb{Z}^n$ such that :

- $\tilde{x} = x_1 + \dots + x_k$

- $Ax_i \leq b$ (Baum and Trotter, 1979)

- **Also in more general cases :** cases where **only a submatrix of A is TU** and the others constraints are already aggregated, or somehow “easy to disaggregate”

Example of exact agregation

In some particular cases, agregation is an exact reformulation

→ it is for example the case for what we call
the Min-up/min-down Unit Commitment Problem (MUCP)

(B. Knueven, J. Ostrowski, J.P. Watson, 2017)

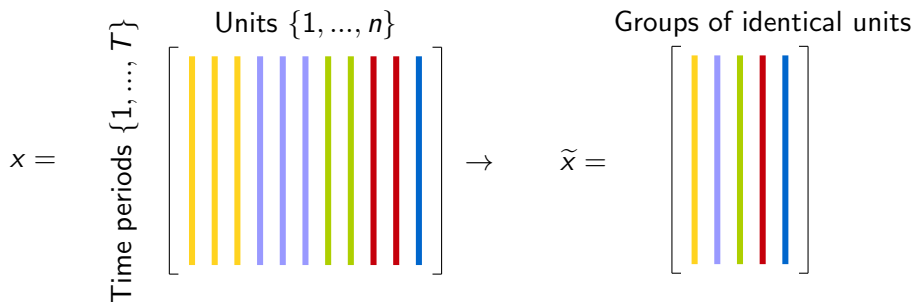
i.e. a restriction of the thermal UCP with only the following constraints :

- **Demand** satisfaction at each time t : $\sum_{i=1}^n p_t^i \geq D_t$
→ **already aggregated**
- **Minimum up and down time** constraints
with formulation from (Rajan, Takriti, 2005)
→ **constraint matrix is TU**
- **Power limits** P_{min} and P_{max} for each unit
once x variables are disaggregated to ensure min-up and down times
→ **easy to disaggregate power variables p**

Example : the MUCP

For a set J of identical units :

$\tilde{x}_{t,J}$ = number of units of J up at time t



$$x_{t,i} \in \{0, 1\}$$

$$\tilde{x}_{t,J} \in \{0, 1, \dots, |J|\}$$

(B. Knueven, J. Ostrowski, J.P. Watson, 2017)

MUCP formulation

$$\begin{aligned}
 v = \min_{x, u, p} \quad & \sum_{t=1}^T \sum_{i=1}^n c_f^i x_t^i + c_p^i p_t^i + c_0^i u_t^i \\
 \text{s. t.} \quad & \sum_{t'=t-L^i+1}^t u_{t'}^i \leq x_t^i & \forall i \in \{1, \dots, n\}, \forall t \in \{L^i, \dots, T\} \\
 & \sum_{t'=t-\ell^i+1}^t u_{t'}^i \leq 1 - x_{t-\ell^i}^i & \forall i \in \{1, \dots, n\}, \forall t \in \{\ell^i + 1, \dots, T\} \\
 & u_t^i \geq x_t^i - x_{t-1}^i & \forall i \in \{1, \dots, n\}, \forall t \in \{2, \dots, T\} \\
 & \sum_{i=1}^n p_t^i \geq D_t & \forall t \in \{1, \dots, T\} \\
 & P_{\min}^i x_t^i \leq p_t^i \leq P_{\max}^i x_t^i & \forall i \in \{1, \dots, n\}, \forall t \in \{1, \dots, T\} \\
 & u_t^i, x_t^i \in \{0, 1\}, p_t^i \in \mathbb{R} & \forall i \in \{1, \dots, n\}, \forall t \in \{1, \dots, T\}
 \end{aligned}$$

Construction of aggregated MUCP formulation

$$\tilde{x}_{t,J} = \sum_{i \in J} x_t^i$$

$$\tilde{u}_{t,J} = \sum_{i \in J} u_t^i$$

$$\tilde{p}_{t,J} = \sum_{i \in J} p_t^i$$

First step : $\sum_{i=1}^n$ replaced by $\sum_{J \in PC} \sum_{i \in J}$

$$v = \min_{x,u,p} \sum_{t=1}^T \sum_{i=1}^n c_t^i x_t^i + c_p^i p_t^i + c_0^i u_t^i$$

$$\text{s. t.} \quad \sum_{t'=t-L^i+1}^t u_{t'}^i \leq x_t^i \quad \forall i \in \{1, \dots, n\}, \forall t \in \{L^i, \dots, T\}$$

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$$\sum_{i=1}^n p_t^i \geq D_t \quad \forall t \in \{1, \dots, T\}$$

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Second step : x_t^i , u_t^i , p_t^i replaced by $\frac{\tilde{x}_{t,J}}{|J|}$, $\frac{\tilde{u}_{t,J}}{|J|}$, $\frac{\tilde{p}_{t,J}}{|J|}$

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Simplify with $|J|$

$$\tilde{v} = \min_{\tilde{x}, \tilde{u}, \tilde{p}} \sum_{t=1}^T \sum_{J \in PC} \sum_{i \in J} c_f^i \frac{\tilde{x}_{t,J}}{|J|} + c_p^i \frac{\tilde{p}_{t,J}}{|J|} + c_0^i \frac{\tilde{u}_{t,J}}{|J|}$$

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Simplify with $|J|$

$$\begin{aligned} \tilde{v} = & \min_{\tilde{x}, \tilde{u}, \tilde{p}} \sum_{t=1}^T \sum_{J \in PC} c_f^i \tilde{x}_{t,J} + c_p^i \tilde{p}_{t,J} + c_0^i \tilde{u}_{t,J} \\ \text{s. t.} \quad & \sum_{t'=t-L^J+1}^t \tilde{u}_{t',J} \leq \tilde{x}_{t,J} & \forall J \in PC, \forall t \in \{L^J, \dots, T\} \\ & \sum_{t'=t-\ell^J+1}^t \tilde{u}_{t',J} \leq |J| - \tilde{x}_{t-\ell^J,J} & \forall J \in PC, \forall t \in \{\ell^J + 1, \dots, T\} \\ & \tilde{u}_{t,J} \geq \tilde{x}_{t,J} - \tilde{x}_{t-1,J} & \forall J \in PC, \forall t \in \{2, \dots, T\} \\ & \sum_{J \in PC} \tilde{p}_{t,J} \geq D_t & \forall t \in \{1, \dots, T\} \\ & P_{\min}^i \tilde{x}_{t,J} \leq \tilde{p}_{t,J} \leq P_{\max}^i \tilde{x}_{t,J} & \forall J \in PC, \forall t \in \{1, \dots, T\} \\ & \tilde{x}_{t,J}, \tilde{u}_{t,J} \in \{0, 1, \dots, |J|\}, \tilde{p}_{t,J} \in \mathbb{R} & \forall J \in PC, \forall t \in \{1, \dots, T\} \end{aligned}$$

Aggregated MUCP formulation

$$\begin{aligned}
 \tilde{v} = \min_{\tilde{x}, \tilde{u}, \tilde{p}} \quad & \sum_{t=1}^T \sum_{J \in PC} c_f^i \tilde{x}_{t,J} + c_p^i \tilde{p}_{t,J} + c_0^i \tilde{u}_{t,J} \\
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 \end{aligned}$$

Here $\tilde{v} = v$: linked to the **total unimodularity** of **blue constraints matrix**.

(Baum and Trotter, 1979)

Unités thermiques “non agrégeables”

Par contre, si on avait :

- **Coûts de variation de puissance**

Variables $\delta > 0$ pénalisées dans l'objectif, telles que :

$$\delta_t^i \geq p_t^i - p_{t-1}^i \quad \delta_t^i \geq p_{t-1}^i - p_t^i$$

Ces contraintes concernent les coûts, mais **n'impactent pas la faisabilité**

On peut agréger dans le sens où il existe une désagrégation faisable
mais la **vision agrégée du coût est inexacte**

→ au sein d'un agrégat, les variations de puissance peuvent se compenser mutuellement

Et si on avait plutôt :

- **Contraintes de gradients**

$$-RD \leq p_t^i - p_{t-1}^i \leq RU$$

Ces contraintes concernent directement la faisabilité.

Il se peut qu'une solution agrégée n'admette pas de solution désagrégée faisable

Exemple : coût de variation de puissance

- 2 unités de production identiques avec : $P_{min} = 200$; $P_{max} = 300$
- Coût variation puissance : 1

→ **Solution agrégée :**

	$t = 1$	$t = 2$	$t = 3$	
#units up	1	2	2	Coût variation puissance = 300
total power	300	400	600	

→ **Solution désagrégée :**

	$t = 1$	$t = 2$	$t = 3$	
unit 1	0	200	300	Coût variation puissance = 500
unit 2	300	200	300	

→ il existe une **solution désagrégée faisable**, mais la **vision agrégée du coût de variation de puissance est inexacte** (minorant)

→ au sein d'un agrégat, les variations de puissance peuvent se compenser mutuellement

Aggregation in practice

For the MUCP :

- **Aggregate** variables corresponding to identical units
- **Solve** the aggregated ILP
- **Disaggregate** the obtained aggregated solution
(a simple algorithm exists)

Very efficient in practice !

→ **used for the INSULAR UCP at EDF**

If **production variation costs** are taken into account :

→ **disaggregation possible.**

→ **but this is heuristic (loss of optimality guarantee)**

If **ramp constraints** are taken into account :

→ **disaggregation no longer possible.**

→ other symmetry-breaking techniques must be used

1 The Unit Commitment Problem

- Formal definition
- Resolution at EDF
- MILP formulation for the UCP
- Lagrangian relaxation and Dantzig-Wolfe reformulation

2 Structured symmetries in Integer Linear Programs

- Symmetry-breaking heuristics
- Restrictions to representatives : using the full orbitope
- Reformulations

3 Sub-symmetries in Integer Linear Programming

- Restrictions to representatives : using the full (sub-)orbitope
- Reformulation

4 Numerical illustration on the Unit Commitment Problem

Sub-symmetries

Let $Q \subseteq \mathcal{X}$ a subset of solutions.

- **Sub-symmetry group $\mathcal{G}_Q$** : symmetry group of $\min\{cx \mid x \in Q\}$.
- **Sub-symmetries** : permutations in $\mathcal{G}_Q$.

In general : Sub-symmetry group contains symmetries
that may not exist in the global symmetry group $\mathcal{G}$

"Structured" sub-symmetry := Sub-symmetry group contains
all column-permutations
for a given submatrix

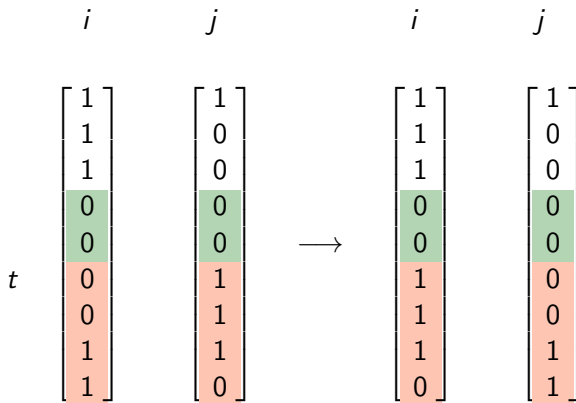
Sub-symmetries in the MUCP

- Example : Two identical units i and j with min-down time $\ell = 2$. Both units are ready to start up at time t

	i	j
	$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$
t		

Sub-symmetries in the MUCP

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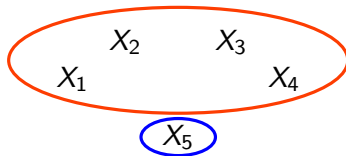


$Q = \{\text{solutions such that } i \text{ and } j \text{ are ready to start up at time } t\}$

Sub-symmetry : Transposition of subcolumns i and j from rows t to T

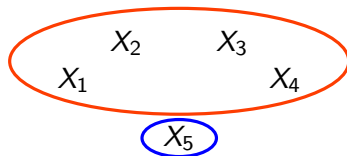
Sub-symmetries – Orbit compatibility

- Recall that global symmetry group $\mathcal{G}$ partitions $\mathcal{X}$ into orbits



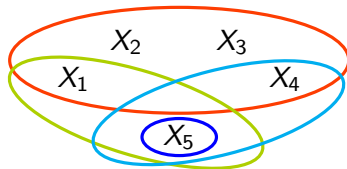
Sub-symmetries – Orbit compatibility

- Recall that global symmetry group $\mathcal{G}$ partitions $\mathcal{X}$ into orbits
- Now consider collection of solution subsets $\mathbb{S} = \{Q_i \subseteq \mathcal{X}, i \in \{1, \dots, s\}\}$ and orbits of groups $\mathcal{G}_{Q_i}, i \in \{1, \dots, s\}$



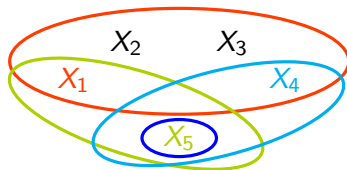
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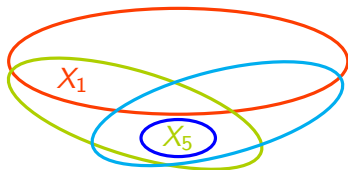
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 - choose one representative
 - apply symmetry-breaking techniques



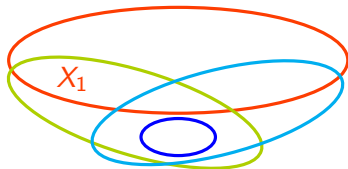
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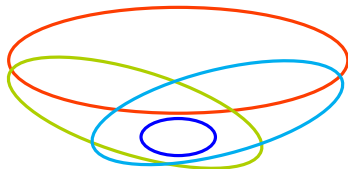
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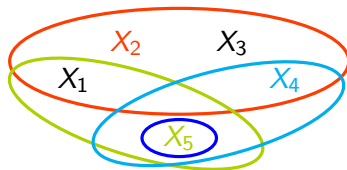
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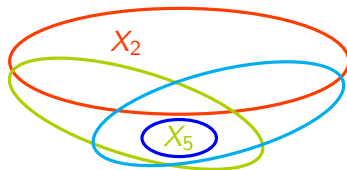
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- “Orbit-compatible”** representatives

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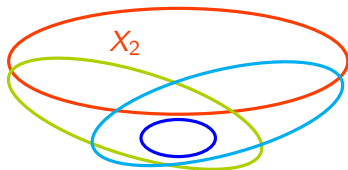
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- “Orbit-compatible”** representatives

Sub-symmetries – Orbit compatibility

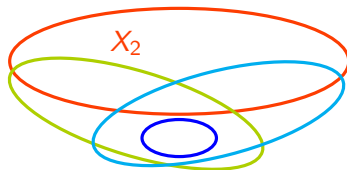
- Recall that global symmetry group $\mathcal{G}$ partitions $\mathcal{X}$ into orbits
- Now consider collection of solution subsets $\mathbb{S} = \{Q_i \subseteq \mathcal{X}, i \in \{1, \dots, s\}\}$ and orbits of groups $\mathcal{G}_{Q_i}, i \in \{1, \dots, s\}$
- Idea : for each orbit of each sub-symmetry group $\mathcal{G}_{Q_i}$:
 - choose one representative
 - apply symmetry-breaking techniques



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Lemma (P.B., P.F., C.R.)

*Lex-max representatives are **orbit-compatible**.*

1 The Unit Commitment Problem

- Formal definition
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2 Structured symmetries in Integer Linear Programs

- Symmetry-breaking heuristics
- Restrictions to representatives : using the full orbitope
- Reformulations

3 Sub-symmetries in Integer Linear Programming

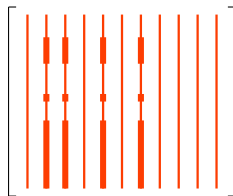
- Restrictions to representatives : using the full (sub-)orbitope
 - Orbitopal fixing
 - Sub-symmetry-breaking inequalities
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4 Numerical illustration on the Unit Commitment Problem

Full sub-orbitopes

Subsets $Q_i \subset \mathcal{X}$, $i \in \{1, \dots, s\}$ such that :

Sub-symmetry group $\mathcal{G}_{Q_i}$ = all column permutations of submatrix (R_i, C_i)



Theorem (P.B., P.F., C.R.)

*Lex-max representatives are **orbit-compatible**.*

Full sub-orbitope $\mathcal{P}_{sub}$

$$\mathcal{P}_{sub} = \text{Conv} \left\{ \text{binary matrices } x \mid \forall i, x \in Q_i \implies x(R_i, C_i) \text{ lex-max} \right\}$$

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Orbitopal fixing for the full sub-orbitope

At each node a of the B&B tree :

- for each Q_i , $i \in \{1, \dots, s\}$
 if all solutions to node a are in Q_i
 apply orbitopal fixing to submatrix (R_i, C_i)

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Collection of solution subsets $\mathbb{S} = \{Q_i \subseteq \mathcal{X}, i \in \{1, \dots, s\}\}$, such that :
 Sub-symmetry group $\mathcal{G}_{Q_i}$ = all column permutations of submatrix (R_i, C_i)

For $i \in \{1, \dots, s\}$, **introduce integer variable** $z_i \geq 0$ such that

$$z_i = 0 \iff x \in Q_i$$

Sub-symmetry-breaking inequality

For columns $c < c' \in C_j$ and row $r_1 = \min(R_j)$:

$$x_{c'}^{r_1} \leq z_i + x_c^{r_1}$$

Diagram illustrating a resource allocation problem. A large bracket on the left groups four vertical bars, each representing a resource. The first two bars are blue and labeled $x_1^{r_1}$ and $x_2^{r_1}$. The next two bars are red and labeled $x_3^{r_1}$ and $x_4^{r_1}$. To the right of the bracket is the label r_1 . Further right is the expression $\in Q_i$. To the right of this is a double-headed arrow $\Longleftrightarrow$ followed by the expression $z_i = 0$. To the right of this is another double-headed arrow $\Longleftrightarrow$ followed by the expression $x_1^{r_1} \geq x_2^{r_1}$.

Sub-symmetry-breaking inequalities

Collection of solution subsets $\mathbb{S} = \{Q_i \subseteq \mathcal{X}, i \in \{1, \dots, s\}\}$, such that :
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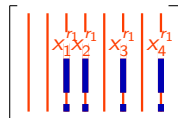
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[illegible]

Full symmetry-breaking condition

Is lexicographic order enforced on remaining rows ?



Full symmetry-breaking condition

Is lexicographic order enforced on remaining rows ?

- Full symmetry-breaking condition ($\mathcal{C}$)

For each $Q_i \in \mathbb{S}$, $\forall k \in R_i$, $\forall c, c' \in C_i$, $\forall x \in Q_i$ s. t. :

$$\left[\begin{array}{c} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{array} \right] \begin{array}{c} c \\ c' \end{array} \begin{array}{c} r_1 \\ k : \text{tie-break} \end{array} = \left[\begin{array}{c} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{array} \right] \begin{array}{c} r_1 \\ k+1 \end{array}$$

$$\exists Q_j \in \mathbb{S} / \left\{ \begin{array}{l} \text{Submatrix } (R_j, C_j) / \\ \text{and } x \in Q_j. \end{array} \right.$$

Satisfying full-symmetry-breaking property

- **It is always possible to reconstruct collection of solution subsets $\mathbb{S} = \{Q_i \subseteq \mathcal{X}, i \in \{1, \dots, s\}\}$ to satisfy full symmetry-breaking condition $(\mathcal{C})$**

Lemma (P.B., P.F., C.R.)

*$\{Q_i, i \in \{1, \dots, s\}\}$ can be reconstructed so that sub-symmetry-breaking inequalities are **full symmetry-breaking**.*

- can be done in polynomial time

These inequalities can be applied to the MUCP, with or without ramp constraints

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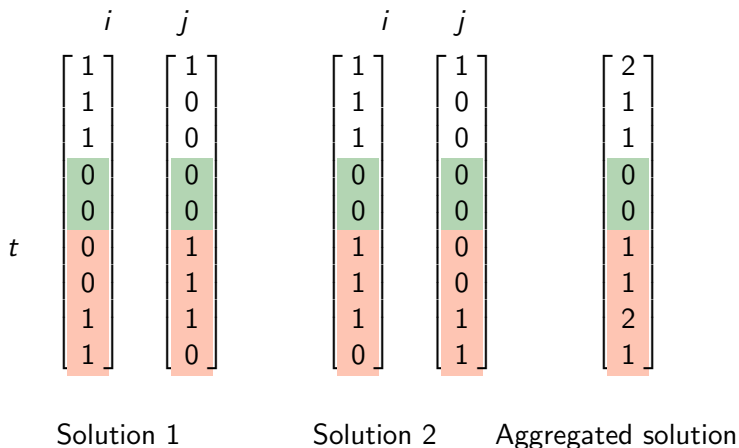
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Aggregation removes also sub-symmetries

- Example : Two identical units i and j with min-down time $\ell = 2$. Both units are ready to start up at time t



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Experimental setting

- **Classical MUCP formulation with or without ramp constraints**

Compared (sub-)symmetry breaking techniques

- Cplex's internal symmetry-breaking techniques
- Aggregation of symmetric variables
- 1-counting inequalities
- Orbital Branching
- Orbitopal fixing for the full (sub-)orbitope
- Full orbitope inequalities
- Full sub-orbitope inequalities

→ **Implemented with Cplex 12.8 C++ API**

Time limit : 3600s

Instances

- MUCP instances
- MUCP instances with ramp constraints.

with sizes $n = 20, 30, 60$ and $T = 48, 96$

Results

- **MUCP instances**

- **Aggregation** is the most efficient technique by far.

Solves MUCP instances in 1 second on average (resp. 100 to 1000 seconds for the other techniques)

- **Ramp-constrained MUCP instances**

- Aggregation not possible anymore.
- When $n > T$ (i.e., $(n, T) = (60, 48)$)
most efficient technique : **full orbitope inequalities**
- When $n < T$
 (i.e., $(n, T) = (20, 48), (20, 96), (30, 48), (30, 96), (60, 96)$)
most efficient technique : **full sub-orbitope inequalities**

Detailed results in *Symmetry-breaking inequalities for ILP with structured sub-symmetries* (Bendotti, Fouilhoux, R)

Some tables : ramp-constrained MUCP

For each size : average on 20 instances.

Size	Technique	#opt	#nodes	gap (%)	CPU (s)
(30,48)	Cplex	4	1 144 442	0.007 19	2922.1
	1-counting Ineq.	6	276 914	0.011 29	2612.0
	Orb. Branch.	1	1 010 438	0.016 57	3480.5
	Fixing Full. Sub-Orb.	7	692 926	0.016 95	2672.5
	Full Orb. Ineq.	9	84 914	0.005 69	2364.2
	Full Sub-Orb. Ineq.	14	346 568	0.002 76	1768.3
(30,96)	Cplex	4	564 244	0.004 83	3009.7
	1-counting Ineq.	4	154 970	0.005 09	3014.7
	Orb. Branch.	1	368 914	0.014 23	3435.3
	Fixing Full Sub-Orb.	2	287 804	0.015 83	3334.7
	Full Orb. Ineq.	5	82 723	0.005 03	2891.4
	Full Sub-Orb. Ineq.	9	205 904	0.003 17	2091.3
(60,48)	Cplex	1	1 190 022	0.003 54	3421.3
	1-counting Ineq.	8	241 259	0.002 17	2424.3
	Orb. Branch.	1	468 546	0.007 40	3433.9
	Fixing Full Sub-Orb.	7	328 643	0.005 02	2626.7
	Full Orb. Ineq.	14	101 805	0.000 35	1751.0
	Full Sub-Orb. Ineq.	8	657 602	0.000 78	2222.8

Conclusion

- **Aggregation is currently used at EDF for insular systems UCP**
 - **Aggregation of identical thermal plants**
 - **Aggregation of identical parallel turbines**
- **Orbitopal fixing for the full orbitope implemented in SCIP !**

(The scip optimization suite 7.0, G. Gamrath et al, 2020)
- **Perspectives :**
 - apply symmetry-breaking techniques to other EDF problems**
 - **Intra-day UCP** (thermal plants feature ramp constraints so aggregation cannot be used)
 - **Smart charging of electric vehicules** (symmetries come from identical vehicules)

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